

The slide features a light blue background with abstract geometric patterns in shades of green and blue along the left and right edges. A red plus sign is located to the left of the title. There are two sets of green 'x' marks: one in the top left and one in the bottom right. A red circle is positioned in the bottom right corner, near the page number.

Phase 1 & 2 Presentation

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TABLE OF CONTENTS

01

Phase 1

You can describe the topic of
the section here

02

Phase 2

You can describe the topic of
the section here



Phase 1

6502

Fabric Utilization Report:

BUF: 322/9720 used (3.3%)

CONB: 3/9720 used (0.0%)

× DFBBP: 143/6480 used (2.2%)

× INV: 360/12960 used (2.8%)

○ NAND: 1228/48600 used (2.5%)

OR: 843/25920 used (3.3%)

aes_128

Fabric Utilization Report:

BUF: 4523/9720 used (46.5%)

CONB: 4/9720 used (0.0%)

× DFBBP: 5018/6480 used (77.4%)

× INV: 6352/12960 used (49.0%)

× NAND: 46283/48600 used (95.2%)

× OR: 23639/25920 used (91.2%)

arith

Fabric Utilization Report:

BUF: 67/9720 used (0.7%)

CONB: 3/9720 used (0.0%)

× DFBBP: 11/6480 used (0.2%)

× INV: 68/12960 used (0.5%)

○ NAND: 187/48600 used (0.4%)

OR: 127/25920 used (0.5%)

z80

Fabric Utilization Report:

BUF: 77/9720 used (0.8%)

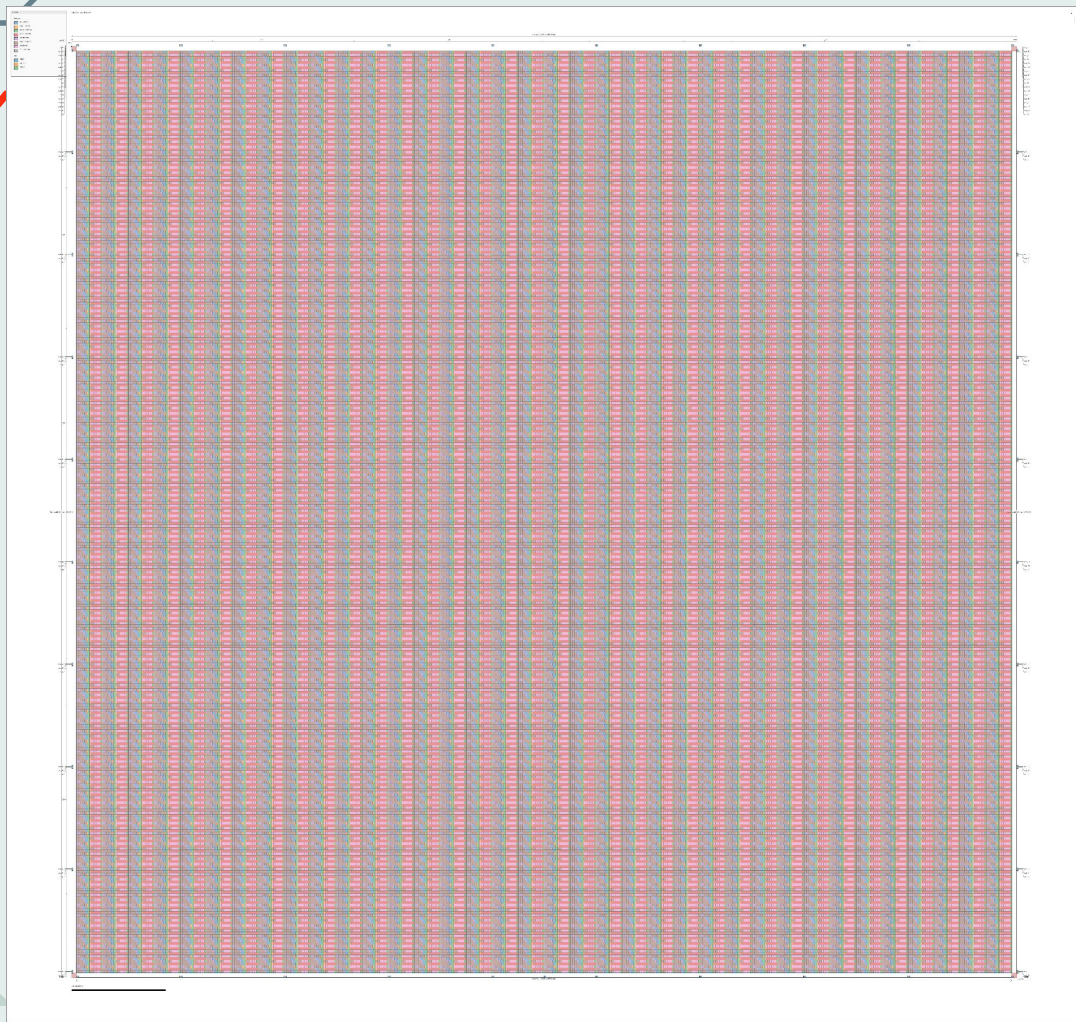
CONB: 3/9720 used (0.0%)

× DFBBP: 160/6480 used (2.5%)

× INV: 1011/12960 used (7.8%)

○ NAND: 3356/48600 used (6.9%)

OR: 4537/25920 used (17.5%)



Phase 2



Greedy Placer Algorithm

Steps

1. Get structures from the .json file and convert specific lists (like occupied_slots) into Sets for fast lookup speed.
2. Identify cells directly connected to fixed I/O pins. Calculate the average position of those pins and place these "Seed" cells in the nearest valid slots to that location.
- ✕ 3. Identify the unplaced neighbors of the Seed cells. Score them based on how many
✕ placed neighbors they have and add them to the queue.
- ✕ 4. Pop the cell with the highest score. Calculate the Barycenter (Center of Gravity) of its
○ already placed neighbors. Find and lock the nearest available slot of the correct type to that calculated center.
5. Once that cell is placed, find its unplaced neighbors. Increment their connectivity scores (because they now have a new "placed" friend) and push/update them in the priority queue. Repeat steps 4-5 until the queue is empty.

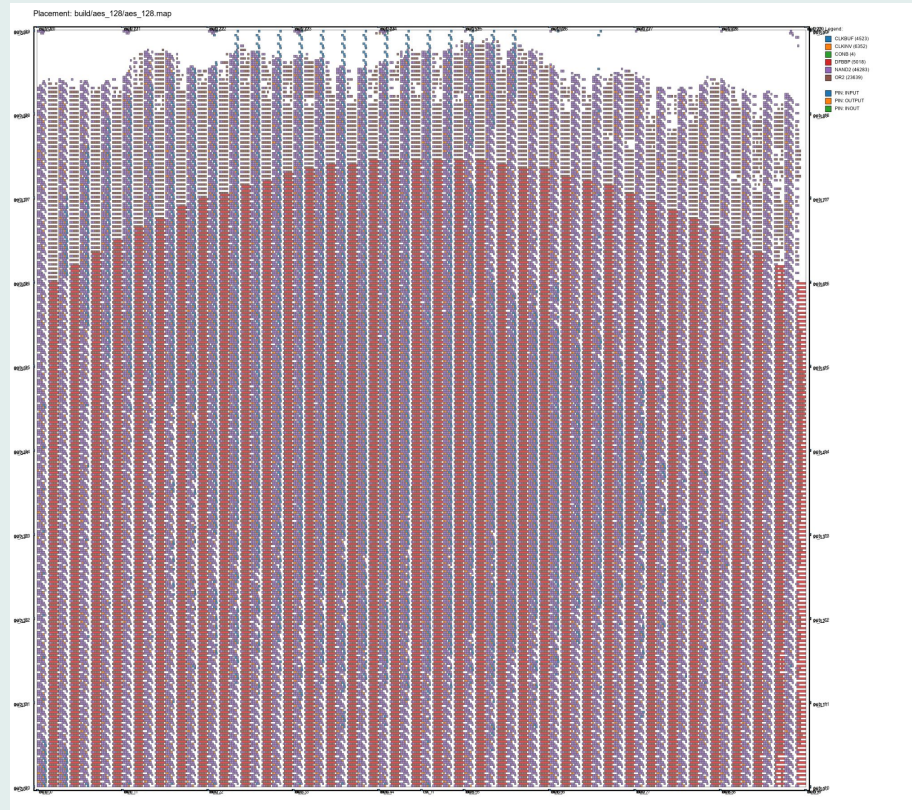


arith

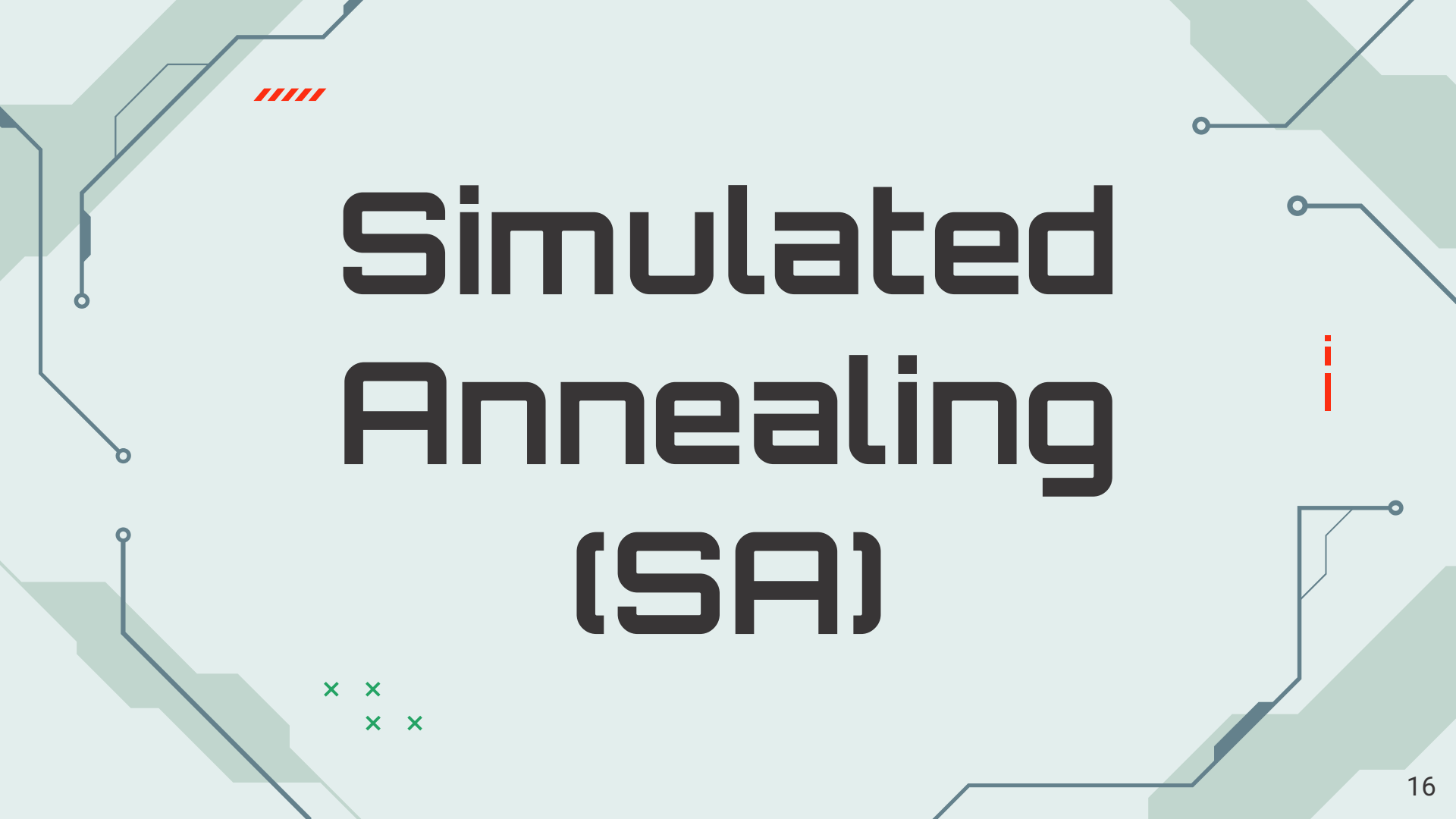




aes_128







Simulated Annealing (SA)

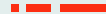
Pseudocode

- 1. Compute `net_to_insts`, `inst_type`, `type_to_insts`, `free_slots_by_type`, 2. `die_bbox`.
- 2. Set `placement = initial_placement`. Compute `current_cost = compute_total_hpw(...)`. Initialize `best_cost = current_cost`, `best_placement = placement`.
- 3. For `k = 0` to `num_temp_steps - 1`:
 - a. Set `T_k`, `W_k` per schedule.
 - b. Repeat `moves_per_temp` times:
 - i. With probability `P_refine`, attempt a swap move; else attempt a `move_to_free_slot`. If that move type is impossible, fall back to the other.
 - ii. After proposal, compute `new_cost`.
 - iii. Apply acceptance rule: if accepted then update `current_cost`, maybe update `best_cost/best_placement`; if rejected then revert.

x
x x
x

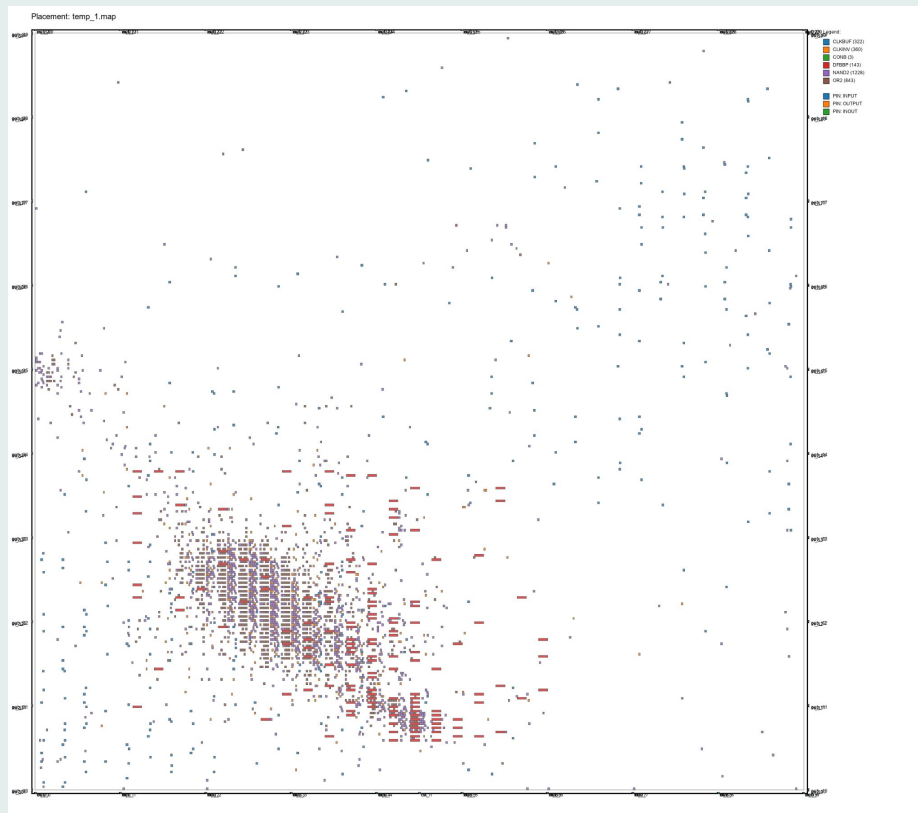
Pseudocode

- 4. After all steps, return best_placement and best_cost.
- 5. Write the best map to output file.





6502

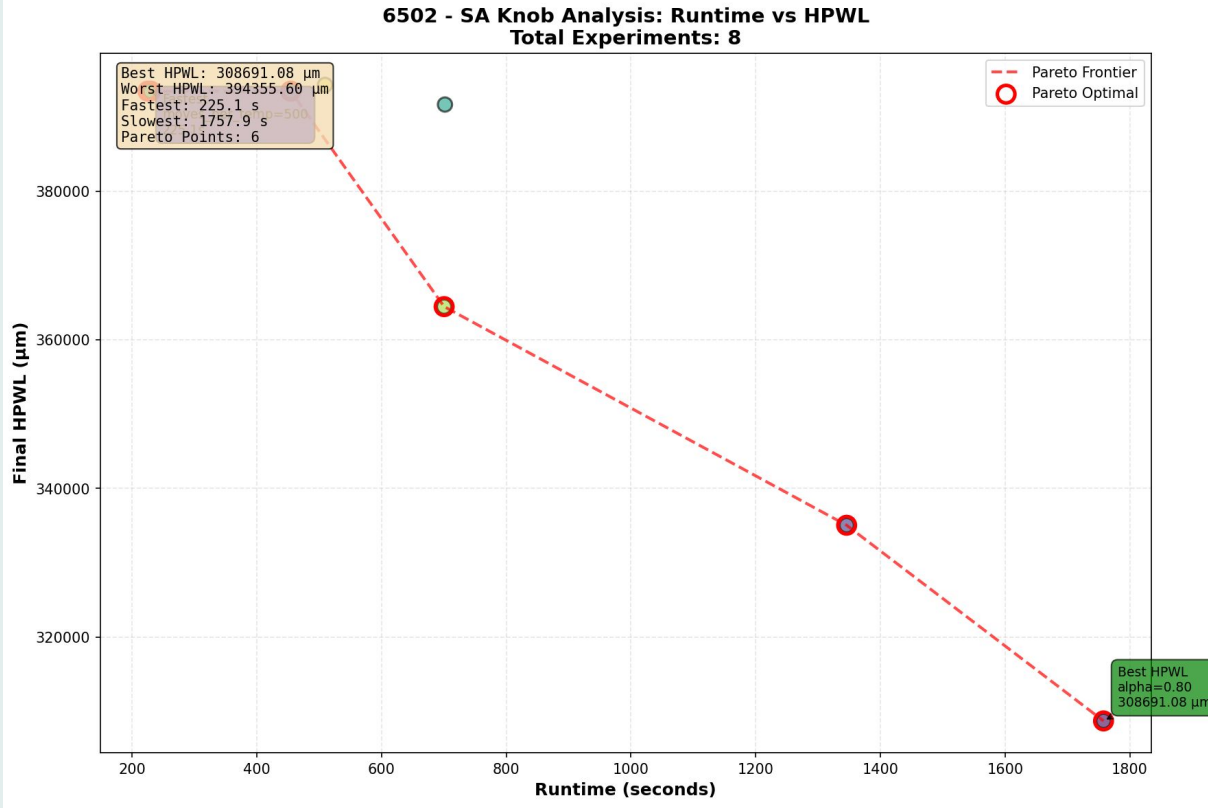




SA-knob Analysis

SA-knob Analysis

Best Parameters: $\alpha=0.80$,
moves_per_temp=1000, $T_{\text{initial}}=200$,
num_temp_steps=60

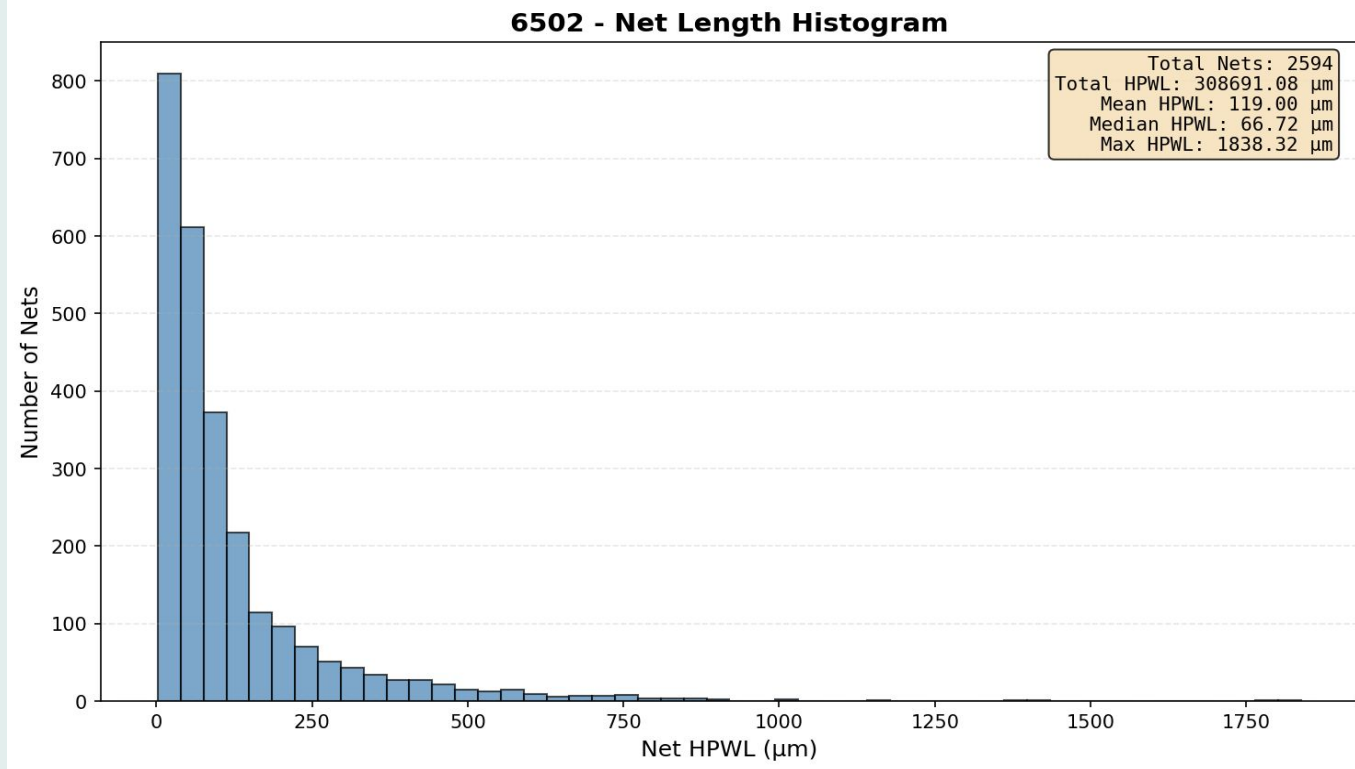




Simulated Annealing (SA- optimized)

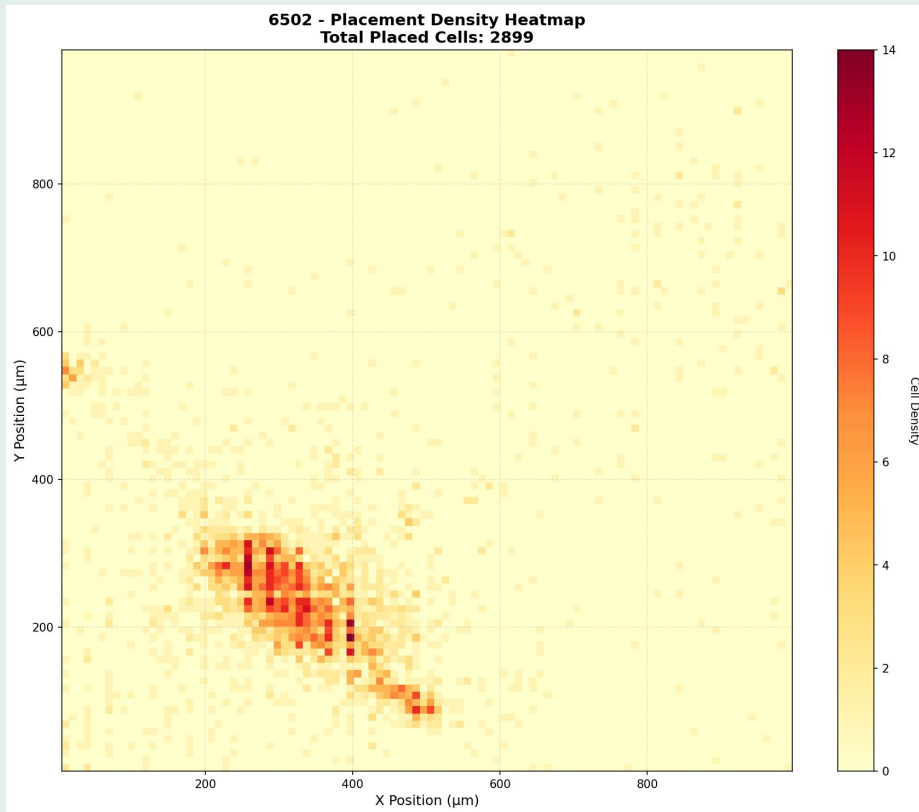


6502 Netlist histogram





6502 placement heatmap



6502 fabric

